



Regulators of mitochondrial dynamics in cancer

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Mitochondrial dynamics encompasses processes associated with mitochondrial fission and fusion, affecting their number, degree of biogenesis, and the induction of mitophagy. These activities determine the balance between mitochondrial energy production and cell death programs. Processes governing mitochondrial dynamics are tightly controlled in physiological conditions and are often deregulated in cancer. Mitochondrial protein homeostasis, transcriptional regulation, and post-translational modification are among processes that govern the control of mitochondrial dynamics. Cancer cells alter mitochondrial dynamics to resist apoptosis and adjust their bioenergetic and biosynthetic needs to support tumor initiating and transformation properties including proliferation, migration, and therapeutic resistance. This review focuses on key regulators of mitochondrial dynamics and their role in cancer.

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Mitochondrial dynamics: regulation and physiological relevance

Mitochondria are dynamic organelles that form filamentous networks or appear as fragmented, rounded structures within cells. These morphologies continuously change as a result of coordinated fission (fragmentation), fusion (elongation), or movement along microtubular structures [1,2]. Fusion and fission are highly conserved processes, orchestrated by dynamin related GTPases: in mammals, cytosolic dynamin related protein (Drp1), is recruited to mitochondria and executes mitochondrial fragmentation (fission) after undergoing extensive post-translational modification (Table 1). Drp1 binds to receptor/adaptor proteins (such as Fis1, MFF, MiD49, and MiD51), at the mitochondrial outer membrane (MOM), and oligomerizes to form a ring-like structure along the

MOM that divides mitochondria by constriction. Such sites are marked by tubules from the endoplasmic reticulum (ER) that ‘pre-constrict’ mitochondria before Drp1 recruitment [1,2]. Two GTPases, mitofusin 1 and 2 (MFN1/2), reportedly mediate MOM fusion, whereas the inner mitochondrial membrane (MIM), is subsequently fused by a process involving the cristae-shaping protein OPA1. Fusion machinery proteins are also subject to posttranslational modifications that regulate their abundance and activity (Table 1).

Although the molecular machinery that shapes mitochondrial morphology is well described, its relationship to mitochondrial functions (such as ATP generation, amino acid and lipid biosynthesis and breakdown, ROS generation, and Ca^{2+} -signaling) is still under evaluation (Figure 1). Mitochondrial shape is particularly plastic during cell cycle progression. Highly connected mitochondria seen in G1/S, are thought to ensure sufficient ATP production during energy-consuming cell proliferation [3], whereas fission increases during S/G2/M as a means to distribute mitochondria equally among daughter cells [4]. In post-mitotic cells, extensive mitochondrial fragmentation occurs during apoptosis [1,2], and the mitochondrial network responds to changes in nutrient or oxygen supply, linking mitochondrial dynamics to cellular signaling pathways and stress responses [5,6,7,8**]. Linked to their role in mitochondrial morphology and function, regulators of mitochondrial dynamics are associated with cytoskeletal proteins and with other cellular organelles. For example, MFN2 tethers mitochondria to the ER and modulates lipid metabolism, calcium homeostasis, and the ER stress response [9]. Mitofusins also link mitochondria to kinesin motors by direct interaction with Miro/Milton, thereby influencing mitochondrial transport [1].

Genetic inactivation of any core mitochondrial-shaping protein promotes embryonic lethality in mice, indicating that mitochondrial dynamics are indispensable for life [10–12]. Whether the perturbation of mitochondrial dynamics per se, or some of the altered processes associated with these dynamics are the cause for the lethality, remain to be determined. In humans, mutations in core proteins have been reported and are associated with tissue-restricted diseases: for example, optic atrophy, in which the loss of retinal ganglions and optic nerve degeneration is linked to OPA1 mutations; Charcot-Marie-Tooth disease type 2A, which is characterized by axonal degeneration of peripheral nerves, results from MFN2 mutations. These observations suggest that altered mitochondrial dynamics and integrity has distinct implications in different cell

Table 1**Post-translational modifications of mammalian mitochondrial dynamics proteins**

Substrate	Regulator	Signal	Effect	Comment	Refs
Drp1	CDK1	P	Activation	Induces fission during mitosis; Aurora A kinase mediates Drp1 recruitment to mitochondria via phosphorylation of RALA.	[4,66]
	CDK5	P	Activation	Increases fission; important for maintenance of glioblastoma stem-like cells	[43**]
	ERK1/2	P	Activation	RAS ^{G12V} -mediated transformation depends on Drp1 activity. Drp1 is essential for MAPK-driven tumors	[8**,38**]
	PKC δ	P	Activation	Increases fission upon oxidative stress; contributes to encephalopathy in rats	[67]
	PKA	P	Inactivation	Limits mitophagy under nutrient starvation; Inhibited under hypoxia to increase apoptosis in cardiomyocytes	[5,26,68]
	CAMKI	P	Activation	Increases Drp1 recruitment and association with Fis1	[69]
	CAMKII	P	Inactivation	Inhibits Drp1 activity; indicative for differentiation in glioblastoma	[43**]
	Rock1	P	Activation	Induces fission under hyperglycemia	[70]
	Calcineurin	De-P	Activation	Reverses PKA-mediated inhibitory phosphorylation	[68]
	NO	S-nitro	Activation	Increases fission in response to β -amyloid; contributes to neuronal damage; controversial	[71,72]
	O-GlcNAc transferase	O-GlcNAc	Activation	Activates fission under high glucose; co-incides with decrease in S637 inhibitory phosphorylation	[73]
	MAPL	SUMO	Activation	Bax/Bak dependent, stabilizes ER-mitochondria contact, required for efficient cytochrome c release	[32**]
	SENP5	De-SUMO	Inactivation/stabilization	Decreases fission and ROS; increases fission during mitosis by stabilizing oligomerization	[74,75]
	SENP3	De-SUMO	Activation	Increased cytochrome c release under neuronal ischemia/reperfusion	[76]
	March5	Ubi	Degradation	Mediates proteasomal degradation of Drp1; essential for Drp1-mediated fission	[56,59]
MFN1	Parkin	Ubi	Degradation	Mediates proteasomal degradation of Drp1; implications in Parkinsons disease	[44]
	APC/C	Ubi	Degradation	Mediates proteasomal degradation of Drp1 during M/G1 transition	[77]
	ERK	P	Inactivation	Inhibits fusion, increases apoptosis through BAK activation	[78]
MFN2	HDAC6	De-AC	Activation	Increases fusion actively under glucose deprivation; more severe starvation increases fusion passively by PKA-mediated inhibition of Drp1	[79*]
	March5	Ubi	Degradation	Targets acetylated Mfn1 more efficiently; prevents Mfn1 accumulation and cell death under mitochondrial stress conditions	[58]
	PINK1	P	Signaling	Fusion-independent role, serves as Parkin receptor at mitochondria	[18]
MFN1/2	JNK	P	Signaling	Marks MFN2 for HUWE1-dependent ubiquitination; inhibits fusion under stress conditions; senses shortage of stearic acid	[80,81**]
	HUWE1	Ubi	Degradation	Marks phosphorylated MFN2 for degradation; inhibits fusion under stress conditions	[80,81**]
	March5	Ubi	Activation	Fusion-independent role; increases mitochondria-ER tethering	[64**]
MFN1/2	Parkin	Ubi	Degradation	Prevents fusion to isolate mitochondria for mitophagy	[82,83]
	USP30	De- Ubi	Inactivation	Limits fusion without altering protein abundance. Similar mechanisms reported in yeast: SCF ^{Mdm30} activates fusion by ubiquitination of Fzo1, which is counteracted by UBP12	[24*,84]

P: Phosphorylation; **De-P:** Dephosphorylation; **S-nitro:** S-nitrosylation; **O-GlcNAc:** O-GlcNAcylation; **SUMO:** SUMOylation; **De-SUMO:** De-SUMOylation; **De-AC:** De-Acetylation;

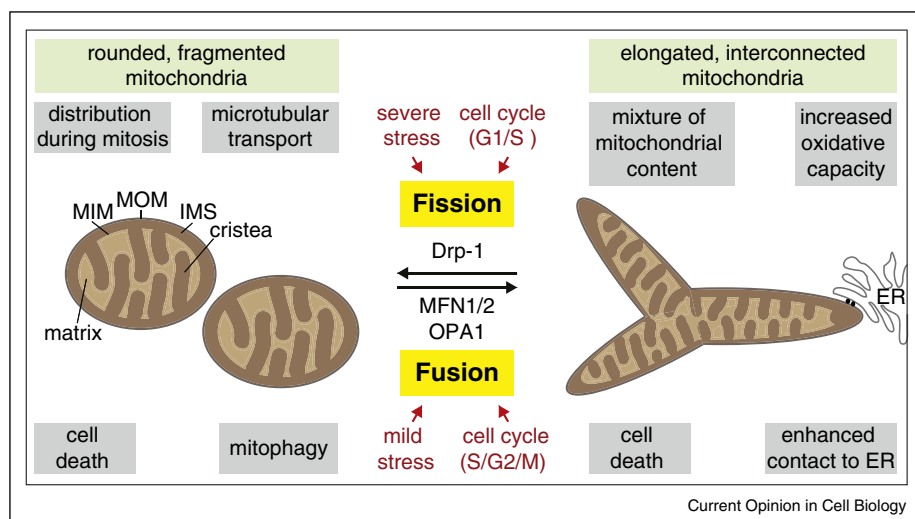
types. Accordingly, perturbations in the balance between fusion and fission have been linked to neurodegenerative and cardiovascular diseases [13,14], and its implication in cancer will be discussed here.

Mitochondrial dynamics, mitochondrial quality control and cell fate decision

Given the importance of mitochondria for vital processes, several mechanisms serve to maintain their integrity [15]. First, accumulation of damaged proteins in mitochondrial compartments causes transcriptional upregulation of

proteases and chaperones, which restore homeostasis by clearing aberrant proteins from the organelle. Additionally, ubiquitin-dependent proteasomal degradation of mitochondrial proteins contributes to mitochondrial quality control (QC). Third, mitochondria-derived vesicles have been demonstrated to carry specific cargo and fuse with the lysosome, providing an additional route to remove molecules from mitochondria [15]. Failure of any of these pathways to maintain homeostasis, results in elimination of the entire mitochondria via mitophagy, or, if stress is sustained, in apoptosis [2].

Figure 1



Mitochondrial Dynamics.

The double membrane structure of the mitochondrion creates distinct compartments. The mitochondrial outer membrane (MOM) and the mitochondrial inner membrane (MIM) separate the intermembrane space (IMS) from the cytoplasm and the mitochondrial matrix. The electron transport machinery is located in invaginations of the MIM, called cristae. Mitochondrial morphology is determined by a balance between fusion and fission events, mediated by dynamin related GTPases (Drp1, MFN1/2, OPA1). Increased fission events or decreased fusion events results in a fragmented mitochondrial network as observed during mitosis or upon severe stress (e.g. hypoxia, apoptotic insults). Increased fission is necessary to equally distribute mitochondria among daughter cells, and mitochondrial trafficking within the cell. It facilitates mitophagy and is associated with apoptosis. When fusion exceeds fission, elongated mitochondria, often highly interconnected, display increased oxidative capacity, and increased contacts with the endoplasmic reticulum (ER). Increased fusion is observed following nutrient starvation or mild stresses and is considered a homeostatic event to dilute damaged mitochondrial molecules (oxidized lipids, proteins, DNA), escape autophagy and prevent cell death, although fusion reportedly enhances cell death in response to certain stimuli.

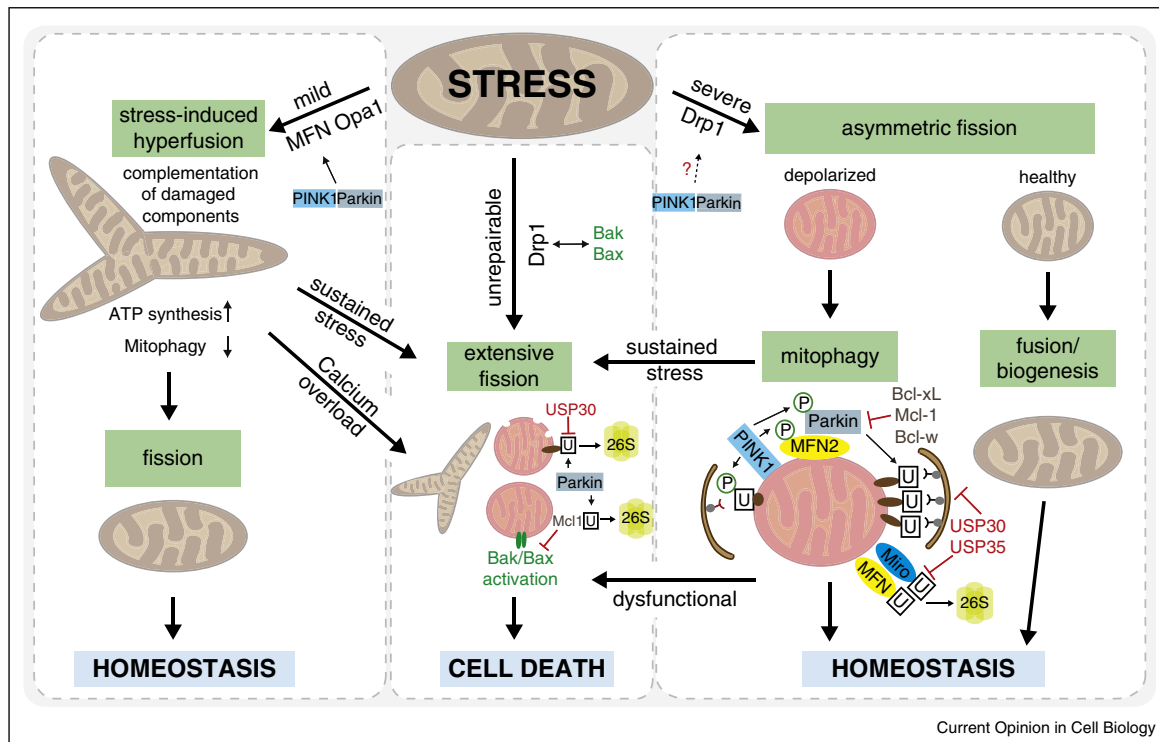
Mitochondrial quality control and cell fate decisions

Among the best-studied regulators of mitochondrial QC, are the ubiquitin ligase Parkin, and the serine threonine kinase Pink1 [1]. Under non-stressed conditions, PINK1 is continuously imported into the mitochondrial intermembrane space, and processed by proteases. PINK1 cleavage products are released into the cytosol and degraded by the proteasome. Upon loss of mitochondrial membrane potential, cleavage is abrogated, PINK1 accumulates in the MOM, undergoes autophosphorylation, and subsequently, PINK1 phosphorylates ubiquitin, MFN2, and Parkin [1]. Phosphorylated ubiquitin serves as a recognition signal for autophagy receptors (OPTN and NDP52), and is sufficient to induce mitophagy, independent of Parkin [16^{••}]. Pink1-mediated phosphorylation of both, Parkin and ubiquitin, is required for Parkin E3-ligase activity [17^{••}]. Phosphorylated MFN2 recruits Parkin to depolarized mitochondria, identifying MFN2 as a Parkin receptor [18]. Parkin-mediated ubiquitination of MOM proteins amplifies initial mitophagy signals, and recruits additional autophagy receptors to the damaged organelle [16^{••}]. Parkin-mediated ubiquitination also targets MOM proteins for proteasomal degradation, among which are MFNs, resulting in inhibition of fusion, and consequently, isolation of dysfunctional

mitochondria from the network [19]. Interestingly, the kinesin adaptor Miro is also target of Parkin-mediated degradation, which arrests mitochondrial motility, further separating damaged mitochondria [1].

The PINK1/Parkin axis functions in mitochondrial QC on several levels, and, dependent on the degree of stress, triggers adaptive responses before the entire organelle is removed (Figure 2). Under moderate stress conditions PINK1 and Parkin reportedly regulate generation of mitochondria-derived vesicles to degrade oxidized mitochondrial components [20]. Similarly, low dose treatment with the mitochondrial uncoupler CCCP results in Parkin-mediated fusion, which limits mitochondrial damage and thus mitophagy in neurons [21] (see next paragraph). Higher doses of CCCP reportedly promote Parkin recruitment and mitophagy, which is antagonized by Parkin interaction with antiapoptotic Bcl-2 proteins (Bcl-xL, Mcl-1, Bcl-w) [22^{••}]. Given that mitophagy serves as a QC mechanism to remove damaged mitochondria and is thought to prevent cell death, inhibition of mitophagy by antiapoptotic proteins appears counterintuitive, unless that inhibition primes mitochondria for induction of apoptosis. Indeed PINK1/Parkin promote mitochondrial depolarization-induced apoptosis by ubiquitinating and

Figure 2



Mitochondrial dynamics, quality control and cell fate decision.

Dependent on the type and severity of stress, mitochondria undergo either fusion or fission to activate homeostatic pathways or precede cell death. Hyperfused mitochondria relieve mitochondrial stress by complementation of damaged content. They resist mitophagy and increase ATP production to adjust to cellular stresses as nutrient deprivation. ER-mitochondrial Ca^{2+} -transfer increased mitochondrial function, while excessive Ca^{2+} -overload results in cell death programs. Asymmetric fission segregates mitochondrial content in a polarized manner, resulting in one hyperpolarized mitochondria that is removed from the network by mitophagy, and one hypopolarized, which is reintroduced into the healthy mitochondrial pool. Asymmetric fission relies on Drp-1 activity, but the underlying mechanism of polarized segregation of mitochondrial content remains to be determined. Pink1/Parkin-mediated mitophagy is under posttranslational control (proteolytic cleavage, phosphorylation and ubiquitination) and can be counteracted by de-ubiquitinating enzymes or protein-protein interaction between Parkin and antiapoptotic Bcl-2 proteins. Increased Parkin-activity is associated with a shift toward a pro-apoptotic response, as excessive proteasomal degradation of MOM proteins result in mitochondrial membrane disruption and cytochrome c leakage. Parkin-mediated ubiquitination and subsequent degradation of antiapoptotic Mcl-1 increases Bax/Bak-dependent membrane permeabilization and triggers apoptosis.

targeting antiapoptotic Mcl-1 for degradation [23^{••}]. Reciprocal activity between Parkin and the Bcl-2 family proteins represents a regulatory node that shifts homeostatic responses toward cell death when mitochondrial stress is sustained. Parkin-mediated effects are further regulated by de-ubiquitinating enzymes, such as USP30 and USP35, which reportedly de-ubiquitinate Parkin substrates to antagonize mitophagy [24[•]] or Parkin-dependent cell death [25[•]].

Fusion and fission cycles link mitochondrial quality and apoptosis

Mitochondrial fusion and fission are both implicated in mitochondrial QC (Figure 2). Fusion is thought to dilute damaged mitochondrial molecules, such as oxidized lipids or proteins, or mutated mitochondrial DNA [2]. Fusion also reportedly spares mitochondria from autophagy activated under nutrient starvation [6,26].

MFN2-dependent tethering of mitochondria to the ER increases upon fusion, enhancing lipid and calcium transfer between organelles [27], but simultaneously increases sensitivity to calcium-dependent apoptosis [28,29]. Fission in turn is critical for mitophagy by generating one depolarized and one hyperpolarized mitochondrion. This asymmetric fission enables the removal of the depolarized mitochondrion, whereas the hyperpolarized can be reintroduced into the healthy mitochondrial network [30].

The interplay between mitophagy, mitochondrial dynamics, and cell death, serves to maintain mitochondrial integrity in response to stress. As fragmented mitochondria appear in apoptosis, fission is suggested to be an integral step of the apoptotic cascade. Indeed, Drp1 knockout mice display defects in neural tube closure, associated with impaired degree of apoptosis. However, MEFs isolated from these mice respond to apoptotic

signals similarly to control MEFs, although cytochrome c release is delayed [11,12], indicating that the consequence of impaired fission on apoptosis is context-dependent. Instead of a strict association between apoptosis and fission, adjustment of mitochondrial shape reportedly modifies the ability of Bax to permeabilize the MOM [31^{••}]. Whereas extensive fission inhibited ER stress-mediated apoptosis in MFN1-deficient cells, restoration of mitochondrial membrane curvature by pharmacological inhibition of Drp1 facilitated cell death [31^{••}]. This observation is in line with the fact that fission per se does not induce apoptosis, and demonstrates that the activity of pro-apoptotic factors depend on the remodeling of mitochondrial shape to efficiently promote apoptosis [31^{••}]. Similarly, MAPL-dependent SUMOylation of Drp1, stabilizes ER-mitochondrial contacts, and facilitates BAX/BAK-dependent cytochrome c release following apoptotic signals [32^{••}]. Inhibition of MAPL delays apoptosis but does not prevent cell death, further indicating that the remodeling of mitochondrial shape modifies the activity of Bcl-2 proteins and determines apoptosis kinetics.

Similar to the interplay of Bcl-2 family proteins and Parkin, alterations in mitochondrial dynamics also mediate calcium-dependent apoptosis, which is enhanced in highly interconnected mitochondria [28,29]. In this context, Mcl-1S, a Mcl-1 splice variant that displays proapoptotic activity, facilitates Ca²⁺-dependent apoptosis by interfering with Drp1 recruitment to mitochondria, allowing unopposed fusion [28]. Thus, distinct pathways merge on the mitochondrial dynamics machinery to modulate network architecture, which either restores homeostasis, or, depending on the type and degree of damage, execute cell death.

Mitochondrial dynamics and cancer

Cancer cells often exhibit fragmented mitochondria; moreover, high expression or enhanced activation of Drp1 and/or downregulation of MFN2, mediate this phenotype in lung cancer [33], metastatic breast cancer [34^{••}], glioblastoma [35], neuroblastoma [36], colorectal cancer cells [37], pancreatic cancers [38^{••}] and melanoma [8^{••}]. Enhanced fission or reduced fusion is linked to several cancer-related phenotypes, and reversal of this phenotype by Drp1 inhibition or MFN2 overexpression, promotes cell cycle arrest [33] and increased spontaneous apoptosis [33,37]. Mitochondrial fragmentation with a concomitant decrease in complex I activity following survivin overexpression, enhances glycolysis, limits ROS accumulation, and decreases the response to cytotoxic therapies [36]. Although most studies reported fragmented mitochondria in cancer cells, interconnected mitochondria are observed following changes in glucose availability [7], or in response to anticancer therapies such as MAPK- or PI3K-inhibitors [8^{••},38^{••},39^{••}]. These observations are evidence that the dynamics machinery

remains intact in cancer, and could contribute to homeostatic adjustments under oxygen/nutrient shortage, or in response to therapeutic interventions.

Ras-induced transformation depends on fission

Two independent studies demonstrate that MAPK signaling, which is commonly activated in cancer, activates Drp1 via ERK-mediated phosphorylation, and that Drp1 activity is crucial for Ras-driven transformation [8^{••},38^{••}]. Drp1 activity is essential for pancreatic cancer growth in xenograft experiments [38^{••}]. Notably, Drp1 contributes to Ras-induced mitochondrial reprogramming, which is characterized by decreased membrane potential, oxygen consumption, and ATP synthesis, and increased ROS [8^{••}]. In melanoma, the BRAF^{V600E}-mutation correlates with Drp1 phosphorylation, whereas MAPK-inhibition reverses Drp1-mediated mitochondrial fragmentation, improves mitochondrial function, and sensitizes cells to mitochondria-targeting drugs [8^{••}].

Mitochondrial QC and cancer

Besides Pink1/Parkin, several proteins are implicated in control of mitophagy [40]. In brief, NIX, BNIP3 and FUNDC1 are MOM proteins, which serve as autophagy receptors in an ubiquitin-independent manner. Furthermore, the phospholipid cardiolipin can initiate mitophagy. Finally, ubiquitin ligases, such as SMURF1 (although independent of its UBL activity), and MAPL can induce mitophagy. Thus, although Parkin is frequently inactivated genetically in cancer [41], mitophagy can occur in cancer cells lacking Parkin activity, and related phenotypes might be mediated by mitophagy-independent Parkin function.

How might cancer cells benefit from Parkin loss? Since Parkin harbors proapoptotic function, and cancer-associated mitochondrial dysfunction/reprogramming is linked to mitochondrial fragmentation and elevated ROS production, Parkin loss could increase resistance to mitochondria-induced apoptosis. Increased expression of Drp1 and Mcl-1, which are targeted for degradation by Parkin and often overexpressed in cancer, support this idea.

Notably, Drp1-dependent fission reportedly functions in stem cell maintenance in immortalized mammary epithelial stem-like cells [42^{••}]. Labeling of 'old' mitochondrial proteins revealed that, upon asymmetric cell division, stem-like cells contained a greater number of 'new' mitochondria, whereas cells containing more 'aged' mitochondria were less efficient in mammosphere formation and, possibly through increased ROS, primed to differentiate. Interfering with Drp1 or Parkin activity abrogated asymmetric distribution of mitochondria and reduced stem-cell properties *in vitro* [42^{••}]. Although these results await confirmation in adult stem cells *in vivo*, they suggest that mitochondrial fitness regulates

stemness. Interestingly, high Drp1 expression and mitochondrial fragmentation contribute to maintenance of brain tumor-initiating cells (BTICs) [43^{••}]. Although expression of a constitutive-active Drp1 mutant is not sufficient to reprogram non-BTICs into tumor-initiating cells, Drp1 activity is required to maintain their tumorigenic potential and prevent AMPK-driven apoptosis [43^{••}]. As Parkin is inactivated in glioma [41], Parkin-independent mitochondrial QC may contribute to BTIC maintenance. As Parkin reportedly destabilizes Drp1 [44], its loss in cancer cells may contribute to stemness by upregulating Drp1.

Mitochondrial dynamics are implicated in cancer cell migration

Emerging evidence supports a role for mitochondrial dynamics in tumor cell migration in glioblastoma and breast, lung or prostate cancer [34^{••},45–48]. These studies suggest that, similar to activities reported in neurons, mitochondria are trafficked to sites of high-energy demand. In migrating cancer cells, mitochondria accumulate at the leading edge, where processes requiring high energy, as focal adhesion dynamics, occur. Interfering with the mitochondria/microtubule linkage via knock-down of the Rho-GTPase Miro1 inhibits mitochondrial trafficking and reduces cancer cell migration [46]. Most studies report fission as a prerequisite for efficient relocation of mitochondria, and upregulation/activation of Drp1 or downregulation/inactivation of MFN1 is observed in invasive cancer cells [34^{••},45,47,48]. Although Caino and colleagues [39^{••}] observed similar mitochondrial redistribution to the leading edge of migrating glioma cells following PI3K inhibition, mitochondria displayed an elongated phenotype. MFN1 downregulation abolished mitochondrial trafficking and abrogated the migratory response. Thus, mitochondrial dynamics are linked to cancer cell migration, and the relative contribution of fission or fusion appears to be context-dependent. Interestingly, hypoxia is considered a major driver of metabolic reprogramming, tumor progression, and metastasis [49]. We previously established a link between hypoxia and increased fission: Siah2 (a hypoxia-responsive ubiquitin ligase) targets the mitochondrial scaffolding protein AKAP121 for proteasomal degradation, resulting in decreased PKA-dependent inhibitory phosphorylation of Drp1, and thus active fission [5]. Two recent studies link mitochondrial fission to hypoxia-induced migration of breast cancer [50] and glioblastoma [35] cells. Given the importance of hypoxia to tumor progression, further evaluation of the link between mitochondrial dynamics and hypoxia-driven phenotypes is required.

Ubiquitin ligases as emerging regulators of mitochondrial dynamics in cancer

The role of ubiquitin in regulating mitochondrial dynamics and mitochondrial QC has recently been reviewed [51,52], and proteolytic and non-proteolytic functions of

ubiquitin-modifications in mitochondrial dynamics are summarized in Table 1. Above, we summarized Parkin function in regulating mitochondrial dynamics and mitochondrial QC and discussed potential implications of Parkin inactivation in cancer.

Other ubiquitin ligases implicated in mitochondrial dynamics and deregulated in cancer include, for example, Siah2 or March5. Expression and activity of Siah2 is regulated by stress signals often activated in cancer, such as hypoxia, p38 activity, or responses to ER stress [53]. In the absence of Siah2 activity, Drp1/Fis1 interaction is impaired and fission is inhibited. The implication of this regulation for physiological conditions under which fission is required, was demonstrated for cardiomyocyte apoptosis, mice survival to myocardial infarction and *C. elegans* life span [5]. A functional link between protein homeostasis and fission exist as Siah2 is transcriptionally regulated by ER stress, namely the UPR machinery (both ATF4 and sXBP1 bind to Siah2 promoter and induces its transcription) [54]. It will be interesting to evaluate whether Siah2 contributes to mitochondrial fragmentation or mitochondrial reprogramming in cancer.

Studies on the functional role of the mitochondria-resident E3 ubiquitin ligase March5/Mitol in cancer are required, although March5 RING-domain mutations are reported in patients with endometrial cancer [55[•]]. Assessing a potential link between March5 activity and mitochondrial morphology or function is complicated as experimental inactivation of March5 promotes conflicting phenotypes. For example, March5 knockdown [56] or expression of the March5^{H43W}-RING mutant results in elongated, hyperfused mitochondria [55[•],57], which antagonizes mitochondrial stress-induced apoptosis [57,58]. By contrast, March5 knockout cells displays fragmented mitochondria [59,60[•]], and accordingly, are more sensitive to apoptotic insults [60[•]]. It is feasible that the genetic background, cellular stress level or culture conditions influence March5-activity, and further studies are required to clarify the role of March5 in mitochondrial dynamics.

Additional March5 functions, that indirectly affect mitochondrial morphology and may contribute to cancer associated phenotypes, were reported. March5 is implicated in mitochondrial QC mechanisms by targeting denatured mitochondrial proteins (e.g. mutant SOD1 or polyQ proteins), which would otherwise cause mitochondrial damage, for degradation [59]. As a RING finger ubiquitin ligase, March5 regulates its own protein levels (auto-ubiquitination). Mutations within the RING domain of March5 results in its accumulation, with concomitant mitochondrial stress, which is seen as part of a hyperfused mitochondrial phenotype [55[•]], seen in response to stresses that inhibit protein translation [6,26,61]. Thus, accumulation of the March5 mutant induces homeostatic

stress response pathways to inhibit mitophagy, increase ATP production, and prevent apoptosis. Interestingly, March5-mediated hyperfusion results in MAPL-dependent nuclear factor kappa B activation, which contributes to apoptosis resistance by upregulating antiapoptotic proteins [55[•],62[•]].

Mitochondrial Ca²⁺-overload is a potent proapoptotic signal transferred from the ER to the mitochondria, and several mechanisms are reported that interfere with ER-mitochondria-Ca²⁺-transfer to increase apoptosis resistance in cancer [63]. March5 mediates non-proteolytic ubiquitination of mitochondrial MFN2, leading to MFN2 oligomerization and increased mitochondrial-ER tethering. March5 knockdown diminishes these contacts and decreases mitochondrial Ca²⁺ uptake [64^{••}]. Thus, it is tempting to speculate that cancer-related March5 mutants contribute to apoptosis resistance.

Concluding remarks

Almost one century ago, Otto Warburg concluded from his studies on aerobic glycolysis of cancer cells, that mitochondria are inactivated in cancer. His research significantly enhanced our understanding of tumor biology, and metabolic reprogramming is considered a hallmark of cancer. Extensive research in the field of cancer metabolism over the last decades, revisited Warburg's initial conclusion and demonstrated that mitochondria are active in cancer cells and utilized for biosynthetic processes, as well as energy production [65]. Major cancer related pathways (e.g. MAPK, PI3K/AKT) reprogram mitochondrial function and dynamics, and cancer cells rely on such reprogramming for sustained proliferation, the capacity to metastasize or to resist apoptosis, thus positioning mitochondria as an important regulatory hub for major cancer traits. Yet, understanding the contribution of the mitochondrial shape to mitochondrial function and cancer-related phenotypes is in its infancy. Recent advances in the field identified an intricate interplay between regulators of mitochondrial QC, dynamics, and apoptosis, which dictates cell fate under stress conditions. Tumor cells are constantly exposed to such stresses, and trigger adaptive responses that in turn, fuel tumor growth, metastasis, and therapy resistance. Additionally, microenvironmental stresses impact intratumoral heterogeneity and impose stem-like traits on cancer cells. The recent finding that mitochondrial morphology is heterogeneous and contributes to a stem-like phenotype in glioblastoma, implies that mitochondrial dynamics are central players in this activity. Based on the implication of mitochondrial dynamics in fundamental cancer-related processes, future research will deepen our understanding of tumor biology. Our better understanding of processes that are engaged in the control of mitochondrial dynamics and their significance for tumor cell development and maintenance, allow the considerations of means to alter mitochondrial dynamics in cancer, thereby offering new possible

therapeutic modalities. Ongoing and future studies will define whether such approach is feasible and its impact on cancer, at its different stages.

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- of special interest
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